

Numerical modeling on micro-to-macro evolution of crack network for concrete materials

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ABSTRACT

Large quantities of microcracks existing in quasi-brittle materials like concrete can be graphically referred as crack network. Specifically, the network evolution includes microcrack growth and coalescence, macrocrack formation and propagation, and the final rupture. In this regard, the present work aims to formulate a numerical approach that is able to describe the whole process of crack network evolution for concrete materials. Particularly postulated is the stochastic distribution of microcracks in concrete, which coincides with the physical phenomenon in concrete. Highlighted are the growth and coalescence of microcracks, as well as the formation and propagation of macrocracks. A numerical case is carried out to study the concrete specimens that are subjected to tensile loading. The thorough process of crack network evolution is clearly simulated through the approach. Also investigated are the influence of aggregate volume fraction on concrete strength and ductility. The proposed approach provides a guide reference for concrete engineering practice, as well as modeling the complete process of concrete failure.

1. Introduction

Concrete is quasi-brittle material that is extensively applied in engineering practice. A large quantity of flaws or microcracks are inevitably created at the initial state, which can be substantially attributed to the loss of concrete moisture. These microcracks can be graphically depicted as concrete crack network. Under external load, pre-existing flaws or microcracks in concrete can trigger the growth, coalesce into larger cracks that further propagate, eventually lead to the final failure. The so-called evolution of crack network ranges from microcrack growth, crack coalescence, macrocrack formation and propagation, to the final rupture, which is a progressive, permanent and localized internal changes in the process of concrete failure. Noteworthy is that microcrack growth and coalescence occurs globally in concrete, while the formation and propagation of macrocracks is highly localized. Understanding the mechanism of crack network evolution remains to be one of the most challenging issues in concrete materials and components.

Microcracks mostly emerge in the area of cement matrix-aggregate interfaces. Concrete constituents such as cement paste and aggregate are both brittle and almost linear materials. It is generally postulated that the nonlinear behaviors of concrete materials are mainly caused by

the micro-to-macro crack evolution. Nonlinear constitutive equations of concrete is usually employed to investigate the nonlinear behaviors of concrete materials and structures at macroscopic level [1]. Nevertheless, the phenomenological transition from linear brittle to nonlinear quasi-brittle necessitates more accurate modeling at microscopic or mesoscopic scales.

Lattice modeling approach has been extensively employed to simulate the mechanical behaviors of concrete rupture [2–6]. The main idea of this approach is to convert the physical model of concrete into a beam or truss structure. By introducing the criteria for single truss broking, it could finally simulate the failure process of the ‘structure’ resulting from the accumulation of broken trusses. Though it is relatively easy to implement the lattice modeling approach, there still lacks of physical ground of concrete materials.

Energy-based approaches such as strain energy density has been proved to be flexible and successful in characterizing fracture at various scales [7–9]. For example, the averaged Strain Energy Density (SED) method is extended to the static assessment of notched components at the nanoscale [7]. Nevertheless, one of the most widely applied modelling strategies remain to treat the concrete as a multi-phase continuum [10] by introducing nonlinear constitutive equations to each or some of the phases. These equations are generally derived from theory

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of plasticity or/and continuum damage mechanics (CDM) [11]. Accumulative effects of all the phases, represented in the framework of above continuum theories, contribute to the final concrete rupture. Consequently, the failure process as well as global behaviors of concrete materials may be substantially influenced by the designated constitutive equations. In [12], mesoscopic features within the potentially damaged zones are modeled by adopting cohesive zone method, while the other zones are simulated as macroscopic homogeneous material. This cohesive crack [13,12] or smeared crack [14,15] could also be categorized in this continuum-based modeling strategy. It is generally regarded that continuum-based fracture mechanics breaks down at the nanoscale where the discrete nature of atoms cannot be neglected. Nevertheless, the work in [16] shows that the concept of stress intensity factor is still valid provided that the atoms are modeled. Noteworthy is that a reformulated strain energy density averaged over the fracture process zone has been successfully used to quantify the breakdown of continuum fracture mechanics and to characterize the fracture until a sample of few nanometers in length [17]. These approaches could also be treated as continuum-based modelling strategy. In order to reduce the modeling complexity, the non-continuous gaps induced by cracks are treated as nonlinear materials. These mesoscale methods are analogous in a way that phenomenological constitutive equations for concrete phases are employed. They all come down to the continuum-based models. However, it shall be pointed out that, cement paste and aggregate are both brittle and approximately linear materials that have few damage-plasticity-driven nonlinearity.

Recent efforts have been initiated to investigate microcrack behaviors of concrete materials. These modeling approaches include minimum energy based method [18], peridynamic method [19], beam-particle approach [20], independent cover meshless method [21], as well as 3D discrete crack network [22]. It is noteworthy that a three-dimensional mesoscopic simulation of dynamic tensile fracture of concrete is discussed in [23]. On the mesoscale, concrete is taken as a composite of aggregate, mortar and the interface transition zone (ITZ). The nucleation, coalescence, and propagation of cracks are modelled by pre-insert rate-dependent cohesive elements. It is assumed that concrete microcrack are widely distributed before the initiation of macrocracks. The assumption coincides with the physical basis of concrete materials and thus more reasonable simulated results can be expected. Nevertheless, research related to the evolution of crack network in concrete is rather surprising.

The present work is focused on formulating a numerical modeling that is able to describe the complete process of crack network evolution for concrete materials. The micro-to-macro crack evolution includes microcrack growth and coalescence, macrocrack formation and propagation, as well as the final rupture. It shall be emphasized that microcrack distribution in concrete is postulated to be stochastic, which coincides with the physical phenomenon in concrete. A numerical procedure has been developed to simulate crack propagation behaviors at the concrete aggregate-matrix interface in our previous work [24]. The procedure is stipulated to single aggregate-matrix interface. Based on the conclusions in [24], it is assumed that interface crack can be mapped into idealized crack along the main axis of aggregate, such that modeling on the multiple crack coalescence and propagation in concrete could be substantially simplified, which is the starting point of the present work.

The work presented in this paper is structured as follows: described in Section 2 are the simplified crack model as well as the description of crack network in concrete. Numerical modeling on the crack network evolution is presented in Section 3. Applications of the proposed model are discussed in Section 4, including the tensile rupture and effects of aggregate volume fraction. The main conclusions are summarized in Section 5.

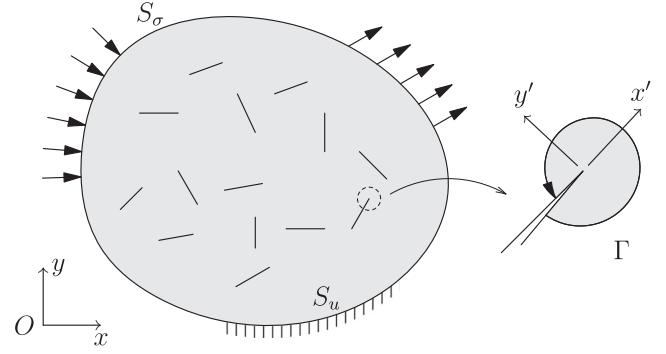


Fig. 1. A continuum containing multiple cracks.

2. Crack network in concrete

2.1. A continuum containing multiple cracks

We consider a body containing multiple cracks that are referred as crack network (Fig. 1), the controlling equations within the area Ω are:

$$\sigma_{ij,j} + f_i = 0 \quad (1)$$

$$\epsilon_{ij} = \frac{1}{2} (u_{i,j} + u_{j,i}) \quad (2)$$

$$\sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij} \quad (3)$$

where $\lambda = \nu E / (1 + \nu)(1 - 2\nu)$ and $\mu = E / 2(1 + \nu)$ are Lamé constants. μ also equals to the shearing modulus G .

On $\Gamma = \partial\Omega = \Gamma_u + \Gamma_\sigma$, boundary conditions are:

$$u_i = \bar{u}_i \quad \text{on } \Gamma_u \quad (4)$$

$$\sigma_{ij} n_j = \bar{T}_i \quad \text{on } \Gamma_\sigma \quad (5)$$

Once the above boundary value problem (BVP) is solved, the stress intensity factors could be extracted. The interaction integrals [25] in the local coordinate systems $x' - y'$ at the crack tip are calculated first (see Fig. 1),

$$J^\alpha = \int_\Gamma \left(\sigma_{ij} \epsilon_{ij}^\alpha n_i - \sigma_{ij} n_i \frac{\partial u_j^\alpha}{\partial x'} - \sigma_{ij}^\alpha n_i \frac{\partial u_j}{\partial x'} \right) d\Gamma \quad (6)$$

here the superscript α represent pure Mode I and Mode II rupture for $\alpha = \text{I}$ and II , respectively. Γ is a contour linking the lower and upper crack surfaces. Accordingly, σ_{ij}^α , ϵ_{ij}^α and u_i^α , which are also expressed in the local $x' - y'$ coordinate systems, are the analytical solutions for the pure Model I or II cracking. Since the interaction integral is path independent, it can generate satisfactory accuracy even with relatively coarse meshes [25]. In order to accurately simulate the stress field at the crack tip, a layer of singular elements is added at crack tips. Details of the mesh can be found in [26]. Using above interaction integrals, the stress intensity factor can be calculated by

$$K_I = \frac{E J^I}{2}, \quad K_{II} = \frac{E J^{II}}{2} \quad (7)$$

For plan strain problems, E and ν should be replaced by $E/(1 - \nu^2)$ and $\nu/(1 - \nu)$, respectively.

2.2. Single microcrack in concrete and its growth behavior

Initial microcracks exist at the interfaces between mortar matrix and aggregates. Since fine aggregates have less influence on crack growing, here we combine fine aggregates and cement pastes together as the matrix of mortar. Also, the interfacial transition zone (ITZ) is not specially modeled for its influences has been considered in the fracture toughness of the mortar/aggregate interfaces.

Fracture toughness of the cement paste, aggregates and the interface have the relationship of $K_{IC,aggregate} \gg K_{IC,cement} > K_{IC,interface}$. Therefore, cracks are not supposed to penetrate into aggregates, meaning this assumption is not suitable for high strength concrete.

Based on above rules, the general failure process could be concluded that with the application of external loads, interfacial microcracks tend to grow initially at the interfaces considering the rule of $K_{IC,cement} > K_{IC,interface}$. Then a large quantity of microcracks may grow into the matrix, merge and/or intersect with others (crack coalescence). Gradually one(or several) localized macrocrack may appear, which may continue to grow with the applied loads, finally lead to the concrete rupture.

Based on the physical mechanism of concrete failure, some cracks may endure a transition from the mortar/aggregate interface into the matrix of mortar. Stresses at interfacial crack tips have a different propriety compared to that of cracks existing in homogeneous materials [27], which makes it difficult to execute the finite element simulation of above transition. This problem may be circumvented by treating all cracks as homogenous ones [28–30].

In this study, we simplify the physical model into a micro-mechanical multiple crack model by generalizing the model proposed by Krajcinovic [30] to the complete rupture process of concrete. Here all cracks are modeled as homogenous ones. For tips locating at mortar-aggregate interfaces, the fracture toughness K_C is set as $K_{C,interface}$, while for those embedded in mortar, K_C is $K_{C,mortar}$ (here $K_{C,interface}$ is smaller than $K_{C,mortar}$). Also, the interfacial segment of the real cracks are mapped along the major axis, as shown in Fig. 2. The real rupture path could be restored by the inverse mapping once the simulation is done.

2.3. Physical and simplified crack model

Fig. 3 shows an example of the physical crack model together with the simplified crack model. Here, displayed in Fig. 3(a) is the physical model. Displayed in Fig. 3(b) is the initial status of crack model where the initial cracks are randomly generated at all interfaces, with random lengths. Fig. 3(c) is the crack configuration before the rupture, which could be mapped back to restore the actual rupture path, as shown in Fig. 3(d).

Theory of linear fracture mechanics is used to control crack growth. After each step, stress intensity factors at all crack tips are calculated and denoted as K_I^J and K_{II}^J . Here I and II represent the modes I (tensile) and II (shearing) fracture, respectively. J indicates the specific crack. The equivalent stress factor is calculated as

$$K_J = K_{eq}^J = \sqrt{(K_I^J)^2 + (K_{II}^J)^2} \quad (8)$$

The criteria for crack growth is

$$\begin{aligned} \text{if } K_J < K_{C,J}: \quad \Delta l_J &= 0 \text{ (no growth)} \\ \text{if } K_J > K_{C,J} \text{ and } K_I^J > 0: \quad \Delta l_J &> 0 \text{ (growth)} \end{aligned} \quad (9)$$

where $K_{C,J}$ is the fracture toughness of the J -th crack tip. It has different values at various conditions of tip location that

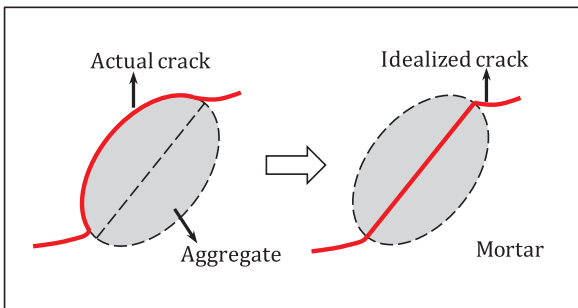


Fig. 2. A diagram of crack simplification.

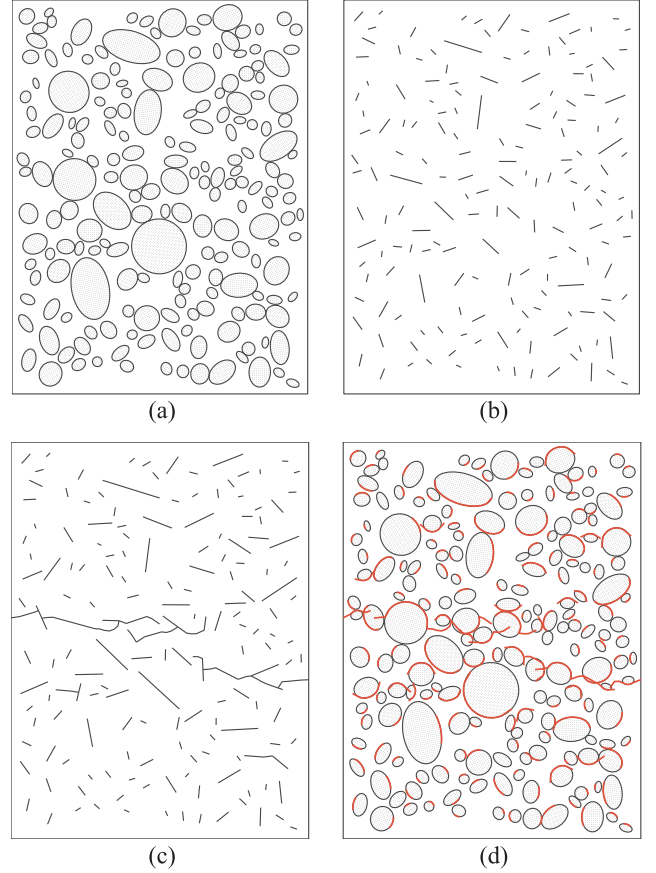


Fig. 3. Physical model and simplified crack model.

1. If at interface: $K_{C,J} = K_{C,interface}$
2. If in mortar: $K_{C,J} = K_{C,mortar}$

In each step, we first apply a tentative load, F_0 or d_0 , depending on the load condition. Solving the problem to obtain the stress intensity factors K_J , we then determine the *critical* tip β that is closest to its fracture toughness, i.e.

$$\beta = \left\{ J: \max_{J=1}^{N_{tips}} \frac{K_J}{K_{C,J}} \right\} \quad (10)$$

where N_{tips} is total number of crack tips.

The load parameter λ is chosen so that the stress intensity factor of tip β equals the fracture toughness $K_{C,\beta}$

$$\lambda = \frac{K_{C,\beta}}{K_\beta} \quad (11)$$

The crack propagation angle at the tip J (the critical tip) is determined based on the following principles

1. If at interface, it will propagate along the interface, and immediately to the two ends of the major axis [24]. In the following steps, the crack will be treated as normal one which has a fracture toughness of $K_{C,mortar}$
2. If in mortar, it will propagate for a length of Δl by default, in the direction of maximum circumferential stress,

$$\theta = \arctan \frac{K_{II}}{\sqrt{K_I^2 + 8K_{II}^2}} - \arctan \frac{3K_{II}}{K_I} \quad (12)$$

The simulation is a K -controlled process that only the cracks with

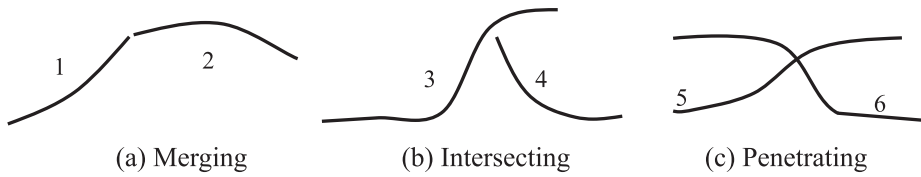


Fig. 4. Basic modes of crack linking.

the maximum of $\frac{K}{K_C}$ grow. Loads are adjusted so as to keep them remaining approximately at the value of fracture toughness. Note the default propagation length is assumed to be the equivalent size of aggregate, which is approximately 2mm.

3. Modeling on crack network evolution

3.1. Basic modes of crack linking

Though shape and distance of cracks also have influence over crack interaction, in the proposed numerical strategy, we are mainly focused on the interaction from the perspective of mathematical modelling. [—] Here the morphology of crack coalescence may be complex. However, it could be decomposed into 3 fundamental modes from which complex morphology could be constructed. As shown in Fig. 4a, the two endpoints of Crack 1 and 2 meet together such that the two cracks merge into a single crack. Fig. 4b shows Crack 4 is growing to intersect with Crack 3. In this situation, stress singularity at the intersected endpoint on Crack 4 vanishes, and the crack will stop growing in this direction. Therefore, the situation of Fig. 4b will not happen in the study even though it could also be simulated by methods developed below.

3.2. Crack merging and intersecting

There are recently some methods for modeling multiple crack growth [31,4,32], most of which are based on some specific FEM techniques, such as XFEM, Voronoi cell finite element method, and so on. Here we develop a method that is based on conventional FEM and could be implemented on the platform of commercial software ABAQUS.

The crack surface, as a directed curve, is stored in the form of a vector containing discrete geometric points. We introduce two statuses — the *locked* and *unlocked* — to describe the singularity on each side (*head* or *tail*) of the crack. The unlocked tip has stress singularity, while the locked tip not. As an example shown in Fig. 5, crack *I* is represented by $n + 1$, indexed from 0 to n , consecutive geometric points. It has an unlocked (marked by a circle) head and a locked (marked by a triangle) tail. Italic capitals, such as *I*, *J*, etc., are used for indexing cracks, and italic lowercases, such as *i* and *j*, are for indexing geometric points on the crack surface.

Based on above definitions, crack merging and intersecting could be modeled. For an ordinary crack embedded in the continuum, the two sides are all unlocked (Fig. 6a). An edge crack could be modeled by setting one side as locked (the tail of crack *J* in Fig. 6b). If a crack merges with another, the merged two tips are set as locked and the rest of two tips still keep unlocked (Fig. 6c). When a crack intersects with another, the intersected point is locked. As shown in Fig. 6d, the head of

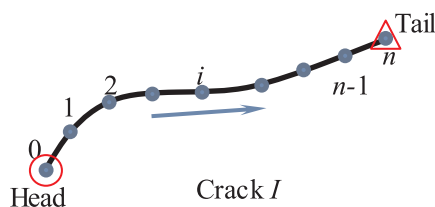


Fig. 5. Representation of a crack.

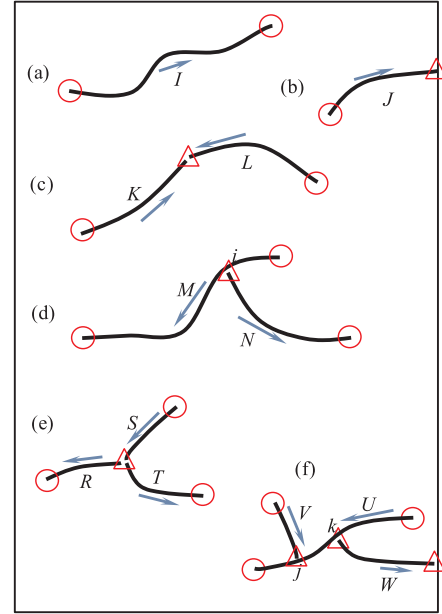


Fig. 6. Several crack geometries in a continuum. (a) ordinary crack; (b) edge crack; (c) crack merging; (d) crack intersecting; (e) 3-crack joint; (f) one crack intersecting with two.

crack *N* is locked because it is growing to intersect with crack *M* at its *i*-th geometric point, while statuses of other 3 tips remain unchanged (here as unlocked). There may have some complex shapes, but all could be constructed through the 4 basic shapes. For an example in Fig. 6e, crack *T* is linked to a point where crack *S* merged with *R* before, which actually becomes a crack intersecting case. Fig. 6f also shows a situation of one crack intersecting with two. Here crack *W* of which one side first reached the boundary intersected with crack *U* at another tip; then *V* grows to intersect with *U*. It could be modeled in a similar way of locking the merged tips. Above two situation both could be simulated in a similar way of locking the merged, intersected and boundary tips. It shall be pointed out that, for the sake of convenience, head and tail are randomly assumed through the numerical process. Head can be locked or also unlocked. And tail remains to be unlocked or locked accordingly.

3.3. Coalescence detection

When a crack is to grow, crack coalescence detection is required to determine if the crack increment will cause a crack interaction or linking. Here, the detection of crack coalescence is discussed together with the detection of crack approaching the boundary edges since similar algorithm is adopted.

In each increment, after the critical tip T_0 on crack *I* as well as its growth direction has been determined by Eq. (10) and (12), a virtual ray along the predicted growth direction is initially created. Then the nearest boundary line, assuming this distance to be $d_{\min}$, could be found and denoted as L_J , where *J* is the index of the line. Here the *boundary lines* include all inner boundaries, outer boundaries and crack surfaces, as the crack could be seen as a general boundary line. If $d_{\min}$ is greater than the default crack increment Δ_{default} , the critical crack will

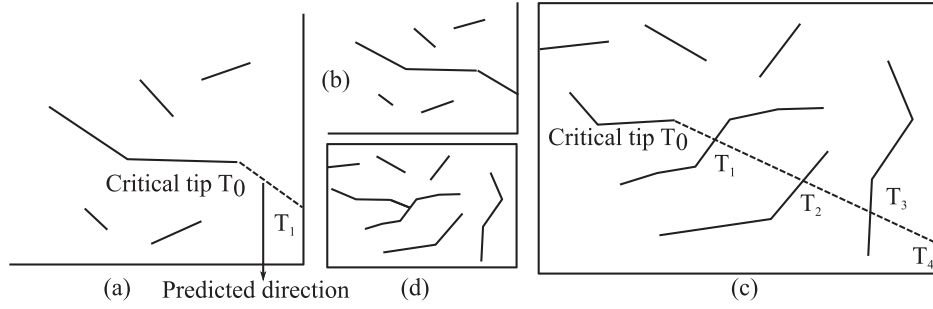


Fig. 7. Detection of crack coalescence.

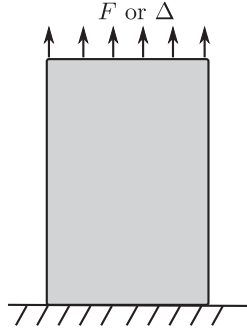


Fig. 8. Tensile specimen.

propagate in the normal way. Otherwise, the crack I will be merged or intersected with the boundary line J , depending on the actual situation. As shown in Fig. 7 is the detection of the crack reaching the boundary edge, while (c) and (d) is the detection of the crack intersecting with a neighborhood.

In the study, a specific module has been developed in Python language to handle these different situations such as detecting crack merging and intersecting, detecting boundary edges, inserting new on-line nodes, treating crack intersection.

4. Applications: tensile rupture and aggregate volume fraction

In this section, based on the simplified crack network model, tensile rupture behavior of concrete specimens is studied, and the influence of aggregate volume fraction on mechanical behaviors is also investigated.

4.1. Aggregate packing

The aggregate mix is first divided into several grades, denoted as $\{G_i: i = 1, 2, \dots, M\}$. Aggregates are seen as eclipses of which the minor-major axis ratio r follows a uniform distribution locating between 0.5 and 1; also they have no preference of the major axis direction. It is also assumed that in grade G_i , the aggregate size (length of major axis) is a random variable uniformly distributing between D_i and D_{i+1} . Therefore for grade G_i , the expected volume value of an aggregates could be calculated as

$$E\left[\pi r D^2\right] = \pi E[r] E[D^2] = 0.75\pi \frac{(D_i^2 + D_i D_{i+1} + D_{i+1}^2)}{3} \quad (13)$$

It could be derived that the representative aggregate size is

$$\bar{D}_i = \sqrt{\frac{(D_i^2 + D_i D_{i+1} + D_{i+1}^2)}{3}}$$

The representative axis ratio $\bar{r}$ is 0.75. Thus by giving the volume proportion V_i of the grade, the number of aggregates in this grade could be estimated as

$$N_i = \frac{V_i}{\pi \bar{r} \bar{D}_i^2}$$

Then n_i aggregates could be sampled and denoted as $\{(d_{ij}, r_{ij}): j = 1, 2, \dots, N_i\}$. Here d_{ij} , the aggregate size, is a random number generated from the uniform distributions with lower and upper bounds of D_i and D_{i+1} , respectively; and r_{ij} , also a generated random number uniformly between 0.5 and 1, is the major-minor axis ratio. After all aggregates $\{(d_{ij}, r_{ij}): i = 1, 2, \dots, M; j = 1, 2, \dots, N_i\}$ are generated, they will be sorted in the descending order of size, and be re-denoted as $\{(d_i, r_i): i = 1, 2, \dots, N\}$. $N = \sum_i N_i$ is the total number of aggregates. Then by giving randomly generated position and main axis direction, these aggregates will be sequentially packed into the container, during which the overlap is not allowed [33,34,10]. If the packing process is successful, a complete information of aggregates could be obtained and written as $\{(x_i, y_i, \theta_i, d_i, r_i): i = 1, 2, \dots, N\}$. Here x_i, y_i and θ_i are x, y coordinates and the direction of the main axis, respectively.

4.2. Case design

All specimens are 150×200 mm cubics subjected to pure axial tension. As show in Fig. 8, the bottom of specimens is fixed while the top side is subjected to uniform force or vertical displacement. Four concrete mixes with different aggregate volume fractions V_a are chosen, which are 15% (denoted as M15, the similar below), 30% (M30), 45% (M45) and 60% (M60), respectively. For each concrete mix, 8–10 specimens are generated. Totally 36 specimens are calculated in the study. Also, all specimens follows the same aggregate grading as shown in Table 1. The proposed numerical plane stress formulation is a case in point here. Plane stress or plane strain case can be adopted based on the real engineering practice.

4.3. Diagram and data analysis

Simulations of the generation and rupture for all specimens follow the same procedure as shown in Fig. 9. Giving the aggregate volume fraction V_a , aggregate packing (Step a) is first conducted to generate the meso-scale heterogeneous structure of the specimen — the physical model. Initial stiffness calculation (Step c) on the physical model is then required for estimating the effective Young's modulus at the ideal *flawless state* of the specimen. After this, the physical model is converted to the multiple crack model (Step d), and by introducing the results of stiffness calculation as an input parameter, the rupture simulation (Step e) on the crack model, starting from multiple crack growth till the final rupture, is performed. During the simulation, the

Table 1
Aggregate grading.

Agg. sieve size (mm)	<32	<24	<16	<12	<8	<5
Percentage	100	94	84	69	54	39

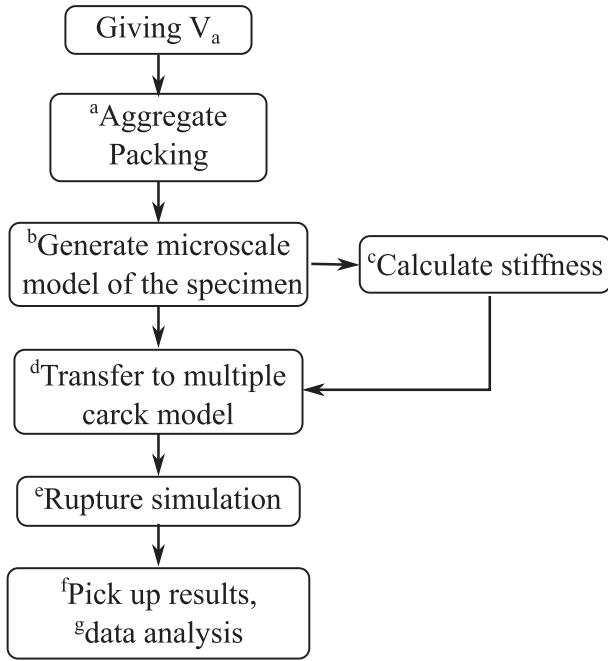


Fig. 9. Computational procedure.

evolutionary history of whole crack network is recorded and outputted in result files, from which necessary information could be extracted (Step f). Also, statistical analysis (Step g) on the results may be needed.

Sub procedures of a, b, d and e have been discussed above. These are general procedures of the model developed in this paper. Here an illustration on procedures c, f, g is given.

Stiffness calculation. Usually, aggregates smaller than 4–5mm are seen as fine aggregates. Here, aggregates of which the sizes are less than 5mm would not be mapped to initial cracks in the procedure of converting the physical model to crack model (Step d). However, the existence of aggregates, including both fine and coarse aggregates, may significantly prompt the stiffness of concrete, that is the reason to perform the Step c: Stiffness calculation.

Fine aggregates, though not been modeled in the crack model, is considered together with cement paste in the form of the mortar (the matrix). The influence of fine aggregates on the Young's modulus is estimated by Hansen's formula [35]

$$E_m = \frac{(1 - V_s)E_p + (1 + V_s)E_s}{(1 + V_s)E_p + (1 - V_s)E_s} E_p \quad (14)$$

where E_m , E_p , E_s are Young's modulus of mortar, paste and aggregates, respectively. V_s is volume fraction of fine aggregates. Here $E_p = 15 \text{ GPa}$, $E_s = 100 \text{ GPa}$ and $V_s = 0.39 V_a$.

Knowing the Young's modulus of mortar E_m , an unit tensile force is applied on the generated physical model to calculate the nominal tensile Young's modulus at the *flawless* state, E_{flawless} . The *flawless* refers to an ideal state that no initial cracks or defects exist inside the concrete specimen, and E_{flawless} is a material parameter needed in the multiple crack model (Step d). Fig. 10 shows an example where (a) is the physical model generated in the aggregate packing process, and (b) is the stress contour of the specimen under unit tensile force.

Extraction of tensile curve. After the rupture simulation is done, necessary information is picked up from result files. Here, we mainly calculate the stress-strain curve for the specimen. The nominal stress σ and ϵ are calculated as

$$\sigma = \frac{F}{WT} \quad (15)$$

and

$$\epsilon = \frac{\Delta l}{H} \quad (16)$$

Here, H , T and T are the width, height and the thickness of the specimen, respectively.

Fig. 11 presents the stress-strain curve. It could be observed that the proposed model can successfully obtain the whole tensile curve, including the linear stage, the pre-peak nonlinearity, the region of post-peak softening, etc. The stress-strain curve calculated from the numerical result is descending for the softening region in a general sense. However, a second peak can be locally observed in the descending curve. It can be attributed to the stochastic phenomenon that the process of crack propagation is temporarily retarded by aggregates randomly distributed in concrete. The phenomenon is also referred as aggregate interlock [36–38]. Nevertheless, the global trend remains to be descending in the softening region. For some specimens with low aggregate volume, post-peak snapback may occur. There couples of parameters could be extracted from the σ - ϵ curve. They are (ϵ_e, σ_e) , (ϵ_p, σ_p) and (ϵ_u, σ_u) , where the subscripts of e, p, and u represent the linear-elastic limit, the stress peak and the ultimate state (rupture), respectively.

Data analysis. Due to the stochastic phenomenon at microstructural scale, actual experimental results on specimens with the same mix may scatter widely. Our numerical results present similar pattern. For each concrete mix in our study, 8–10 specimens are calculated and the maximum and minimum data are not considered in the statistical analysis. Data falling in the medium range are then computed as the representative value of this concrete mix.

Fracture parameters. Our study mainly refers to works by Buyukozturk, etc. [39] for choosing fracture parameters. In the present work, a constantly fracture energy criterion is used, that the critical energy release rate of mortar G_{mortar} is constantly set as 40 J/m^2 . Because specimens may have different Young's modulus, the following expression is used to calculate the fracture toughness of mortar $K_{C, \text{mortar}}$

$$K = \sqrt{EG} \quad (17)$$

Here the Young's modulus E is E_{flawless} . Also, the fracture toughness of the interface is constantly set as $K_{C, \text{interface}} = 1/2 K_{C, \text{mortar}}$ [39,30].

5. Results and discussions

5.1. Concrete failure modes

Figs. 12–15 provides the rupture configurations for some computed specimens (4 cases for each concrete mix, totally 16 cases). It could be seen that the rupture of concrete mainly endures the successive process of microcrack growth and coalescence, macrocrack formation and propagation, and finally the rupture. The proposed model could successfully simulate the complete process.

Although specimen rupture is always caused by the propagation of major cracks, it could be also noticed that there are two modes for the configuration of major cracks. The first mode is that a major crack steadily moves forward by acquiring all microcracks along its growth path. However, sometimes the growth of the major crack is stopped by some aggregates, which may lead to the appearance of the second major crack. At this time, the specimen may finally rupture with two or even more macro eyelets [40], as shown in Fig. 14(a)(b), Fig. 15(a)(b) (c). This is the second mode.

Generally, the second mode preferably occurs in the specimens with high aggregate volume. More aggregates (higher volume fraction) may increase the chance of attracting the major crack. Lower aggregate volume fraction enables the major crack to smoothly move forward. It should be noted that even in high V_a specimens, the first mode still takes the majority lead.

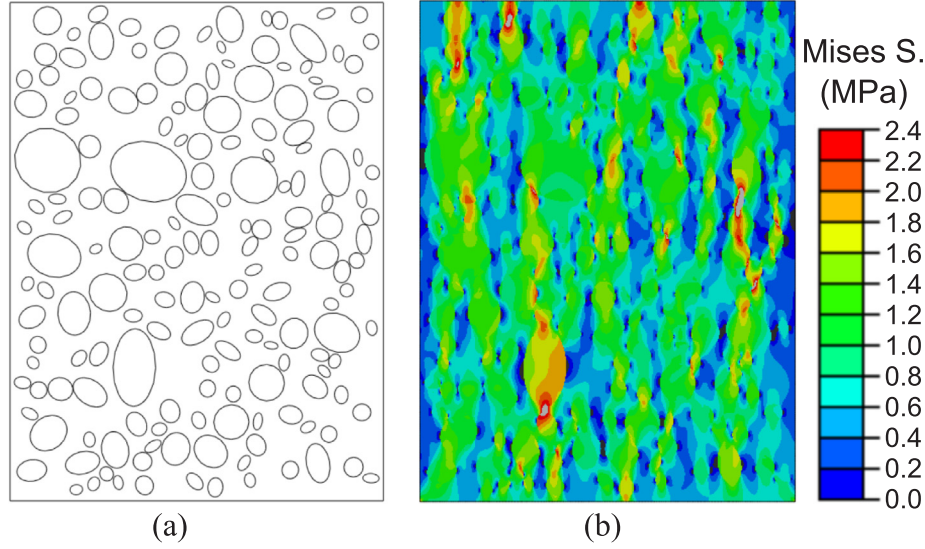


Fig. 10. Stiffness calculation. (a) Physical model (b) stress contour of specimen.

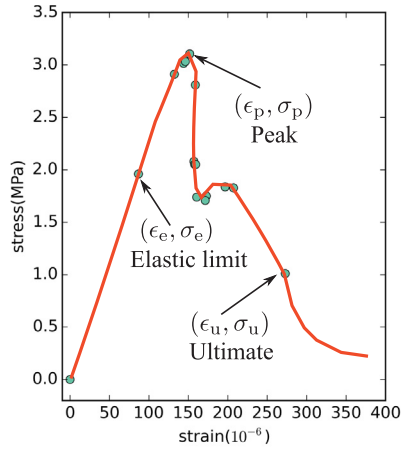


Fig. 11. Stress-strain curve.

5.2. Influence of aggregate volume fraction

In this subsection, the influence of aggregate volume fraction on the concrete performance is numerically investigated. Specifically, parameters below are investigated: initial elastic modulus E_0 , the strength (peak stress) σ_p , the peak strain ϵ_p , ϵ_p/ϵ_e and ϵ_u/ϵ_p .

Elastic modulus. Here the elastic modulus E_0 refers to the initial modulus which takes into account the initially generated microcracks at mortar-aggregate interfaces. It is also the 0-th (initial) status in step d

(Fig. 9) that initial cracks are added into the continuum with the Young's modulus of E_{flawless} . According to mesomechanics theories [41,42], E_{flawless} increases with the increase of aggregate volume fraction. On the other side, more aggregates cause more defects (cracks), which has a negative effect of weakening the concrete stiffness. Results (Fig. 16) show that with the volume fraction increasing from 15% to 60%, E_0 nearly doubles, which agrees with the conclusion in the published literature [43]. It also indicates that comparing to the consolidating effect, the above weakening effect play a minor role in determining the elastic modulus of concretes.

Strength. Displayed in Fig. 17 is the curve of strength versus the concrete mix. It could be observed that the strength generally exhibits a V-shape trend, which corresponds with results in former researches [44,45,36]. This could be described by introducing the theory of non-linear fracture mechanics [36]. However, it is made possible in the numerical case that only linear fracture mechanics is adopted.

Material strength is not necessarily connected to the material stiffness. Stiffness is an overall property that is mainly related to the average of all phases, while strength has a more tendency to be strongly influenced by the short board inside the continuum, such as the inhomogeneity, defect size, etc. In concrete, it was found that the strength is related to the maximum aggregate size [35,39]. With the increase of aggregate volume fraction, the number of coarse aggregates grows correspondingly. Thus, a more critical defect is more likely to appear for defects usually initiated at paste-aggregate interfaces. This may be the reason that the strength decreases when V_a increases from 15% to about 30%.

Fracture mechanics theory can also provide a possible mechanism of

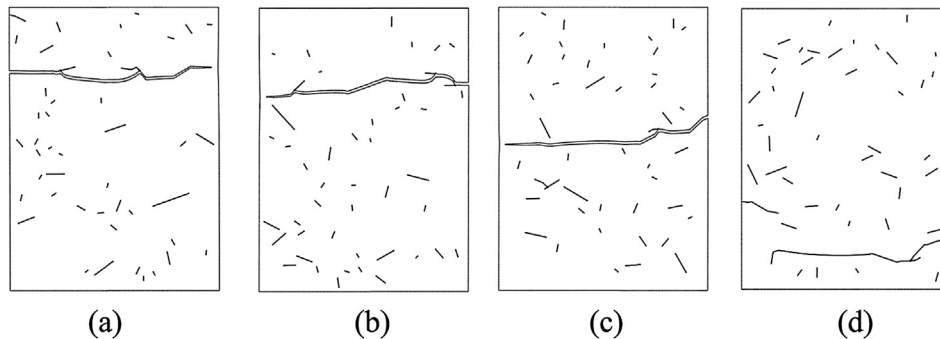


Fig. 12. Failure configurations for M15 specimens.

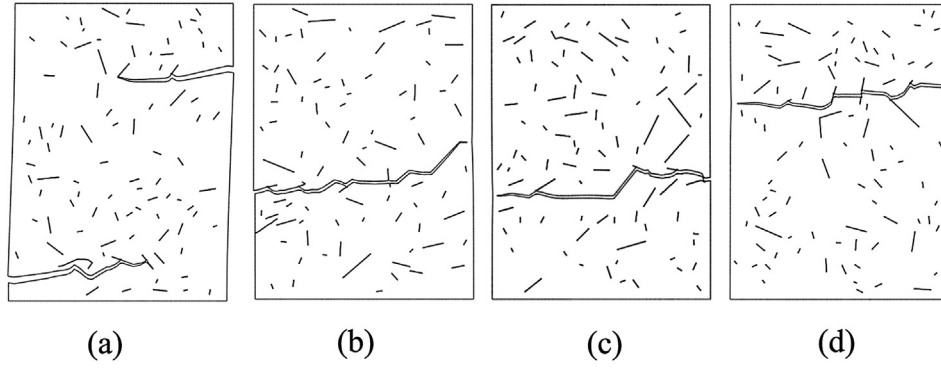


Fig. 13. Failure configurations for M30 specimens.

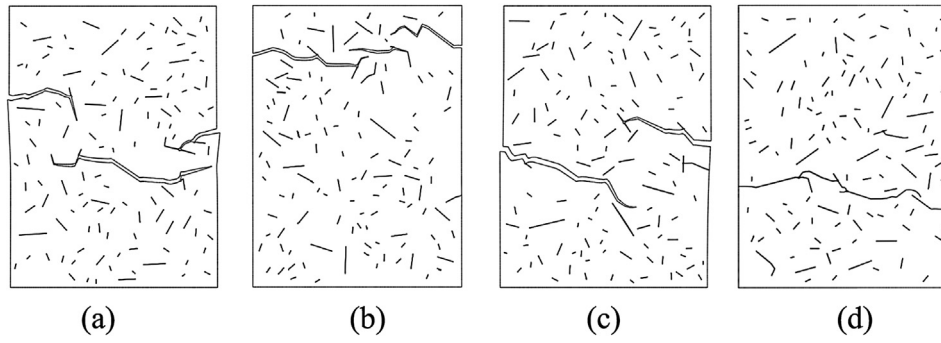


Fig. 14. Failure configurations for M45 specimens.

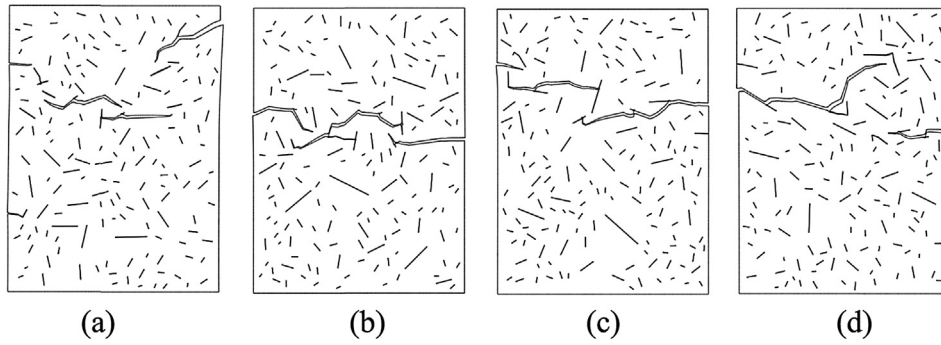


Fig. 15. Failure configurations for M60 specimens.

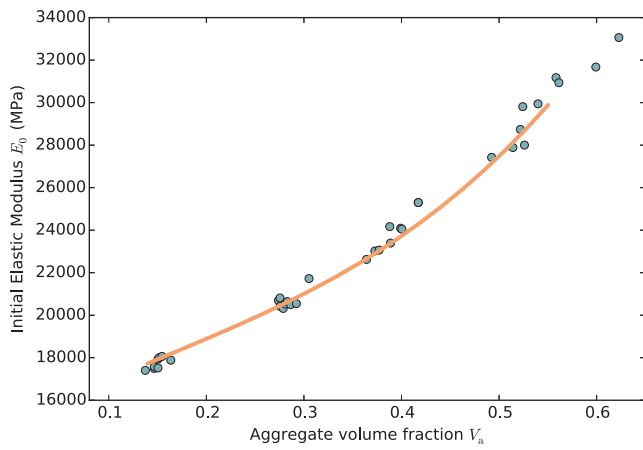


Fig. 16. Initial Elastic Modulus E_0 .

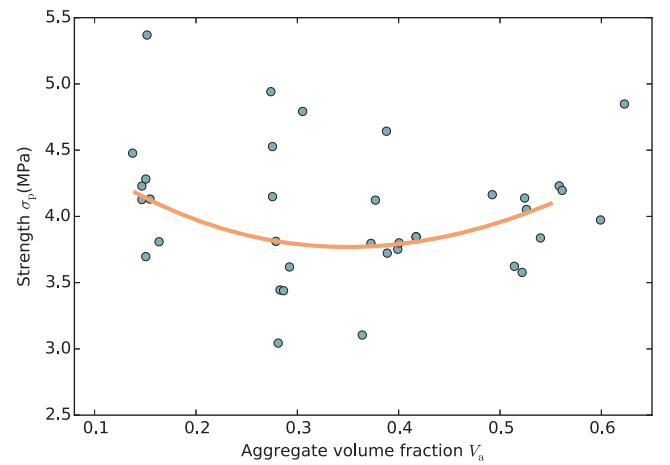


Fig. 17. Peak stress (strength) v.s. aggregate volume fraction.

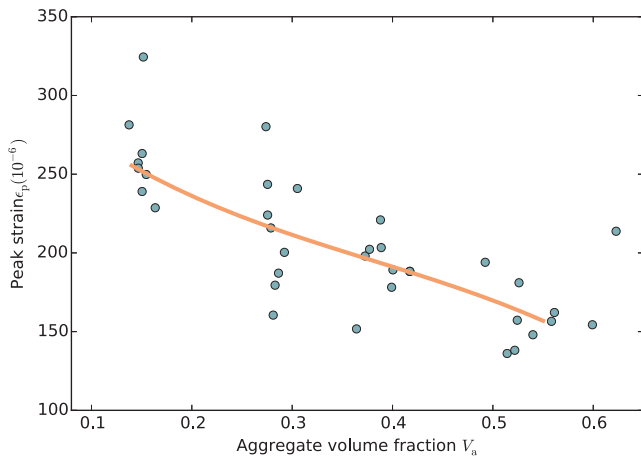


Fig. 18. Peak strain v.s. aggregate volume fraction.

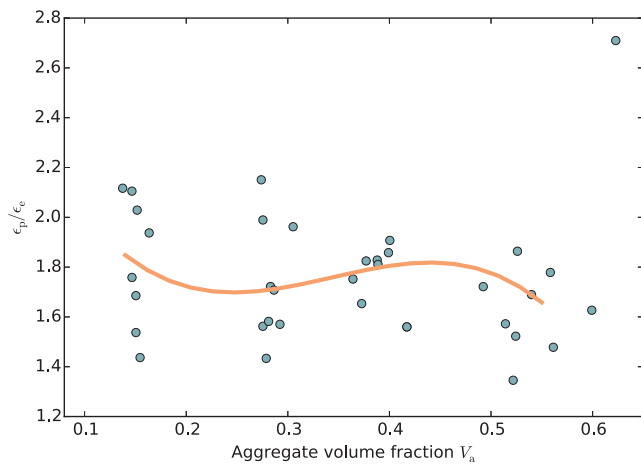


Fig. 19. ϵ_p/ϵ_e v.s. aggregate volume fraction.

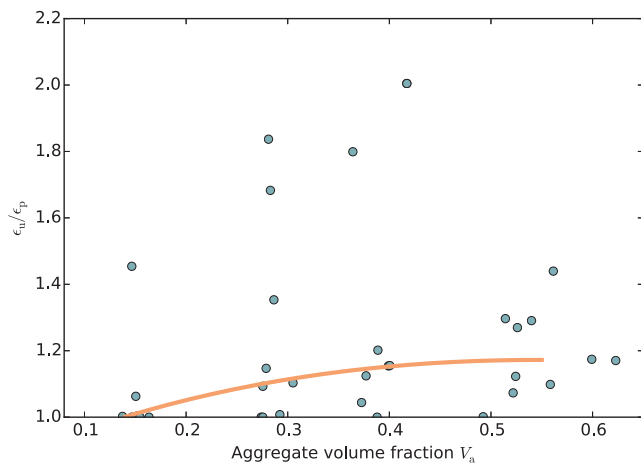


Fig. 20. ϵ_u/ϵ_p v.s. aggregate volume fraction.

stiffness prompting strength in particular cases. From the energy viewpoint, crack starts to grow when the potentially released mechanical energy is sufficient to produce new free surfaces. Since cement paste has a relatively homogeneous meso-structure, the fracture energy could be set to be a constant. From Eq. 17 $K_C = \sqrt{EG_C}$, a greater stiffness E will lead to a greater fracture toughness K_C , if G_C remains constant. And $K_C \propto \sigma_C \sqrt{a}$, meaning that the material strength is accordingly prompted. This may contribute to the strength increase in the 35–45% interval, for the aggregate volume fraction significantly

increase the elastic modulus, as shown Fig. 16.

In a global sense, with the increase of aggregate volume fraction, the scattered initial defects seemly play a major role in weakening the material strength in small volume fraction situations, while in greater volume fraction situations, the increased elastic modulus seemly play a major role in order to protect concrete from rupture.

The peak strain is also plotted in Fig. 18. It constantly decrease with the increase of V_a . This shows that strength enhancement is usually less than the stiffness improvement.

Ductility. ϵ_p/ϵ_e and ϵ_u/ϵ_p (defined in Fig. 11) which is related to the concrete ductility are also investigated. Fig. 19 shows that ϵ_p is usually 1.40–2.2 times of ϵ_e at the hardening stage. However, increasing aggregate volume fraction seemly does not has appreciable influence on the value of ϵ_p/ϵ_e . Fig. 20 shows that the value of ϵ_u/ϵ_p increases together with the increase of aggregate volume fraction. In general, increasing the aggregate volume can help lower the concrete brittleness and improve the ductility.

6. Conclusions

A mesoscale multiple crack model is proposed to investigate the evolution of crack network in concrete materials. Based on our previous work. The assumption is made that each of interfacial crack can be mapped into idealized crack along the main axis of aggregate, such that modelling on the multiple crack coalescence and propagation could be substantially simplified. This approach greatly reduces the computational cost while considering the physically stochastic behavior of microcracks. The whole approach is based on the theory of linear fracture mechanics. Numerical results demonstrate that the proposed model is able to simulate the complete process of concrete rupture, ranging from microcrack growth and coalescence, macrocrack formation and probation, to the final failure. Also made possible is the modeling of crack intersecting, merging as well as crack penetrating boundaries. The numerical modeling is implemented in Abaqus through its Python scripting interface. A specific module in Python language that could handle complex multiple crack morphology is further developed.

The numerical model is applied to tensile rupture modeling of concrete specimens. The influence of aggregate volume fraction on the mechanical behaviors of concrete specimen is investigated. Numerical results show that, the concrete strength exhibits a V-shape trend with the increase of aggregate volume fraction. Increase of the aggregate volume may contribute to lower the concrete brittleness and improve the ductility.

Last but not least, although the numerical model provides a new approach for modeling the failure process of concrete materials, it shall be pointed out that the proposed approach is stipulated to two-dimensional problems at this point. Computational cost will be substantially augmented once three-dimensional problems are treated. Computational efficiency shall be further improved. This will be left for our forthcoming work.

CRedit authorship contribution statement

Baijian Wu: Conceptualization, Methodology, Writing - original draft, Writing - review & editing. **Zhaoxia Li:** Conceptualization, Supervision, Funding acquisition. **Keke Tang:** Conceptualization, Methodology, Writing - review & editing, Funding acquisition.

Declaration of Competing Interest

None.

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